Examination Questions

ELEC50002: Communications, May 2023
Dept. of Electrical and Electronic Engineering, Imperial College London

Name: Solution CID:_____

Instruction:

- This is a closed-book examination. But a standard calculator and three pages (8½" X11") of handwriting notes are allowed.
  - *Standard calculator*: please check the EEE Dept, Imperial College London, for the specifications of a standard calculator.
  - *Handwriting notes*: the notes must be in your original handwriting on three pieces of standard letter size paper (one side only). Please write your name and CID on the top left on each page of the notes.
- Incorrect answers with justification and reasoning can receive partial credit. However, correct answer without justification or reasoning will receive no credit.
- Students need to answer all 6 problems.

Prob I (*14 pts*)	
Prob II (*11 pts*)	
Prob III (*7 pts*)	
Prob IV (*6 pts*)	
Prob V (*10 pts*)	
Prob VI (*12 pts*)	
Total (*60 pts*)	

Information for Students:

Some Fourier Transform Pairs

	$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$	$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$
1	$x(t) = 1$	$X(f) = \delta(f)$
2	$x(t) = \delta(t)$	$X(f) = 1$
3	$x(t) = u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$	$\frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$
4	$x(t) = sgn(t) = \begin{cases} 1, & t \geq 0 \\ -1, & t < 0 \end{cases}$	$X(f) = \frac{1}{j\pi f}$
5	$x(t) = \frac{1}{\pi t}$	$X(f) = -jsgn(f) = \begin{cases} -j, & f \geq 0 \\ j, & f < 0 \end{cases}$
6	$x(t) = rect\left(\frac{t}{T}\right) = \begin{cases} 1, & \lvert t\rvert \leq T/2 \\ 0, & \lvert t\rvert > T/2 \end{cases}$	$X(f) = T\frac{\sin(\pi fT)}{\pi fT} = Tsinc(fT)$
7	$x(t) = W\frac{\sin(\pi Wt)}{\pi Wt} = Wsinc(Wt)$	$X(f) = rect\left(\frac{f}{W}\right) = \begin{cases} 1, & \lvert f\rvert \leq W/2 \\ 0, & \lvert f\rvert > W/2 \end{cases}$
8	$x(t)\cos(2\pi f_o t)$	$\frac{1}{2}X(f - f_0) + \frac{1}{2}X(f + f_0)$
9	$x(t)\sin(2\pi f_o t)$	$\frac{1}{2j}X(f - f_0) - \frac{1}{2j}X(f + f_0)$

Fourier Transform Properties

	Name of Theorem	Function	Fourier Transform
1	Superposition (a_1 and a_2 are arbitrary constants)	$a_1x_1(t) + a_2x_2(t)$	$a_1X_1(f) + a_2X_2(f)$
2	Time delay	$x(t - t_0)$	$X(f)e^{-j2\pi ft_0}$
3	Scale change	$x(at)$	$\lvert a\rvert^{-1}X\left(\frac{f}{a}\right)$
4	Time reversal	$x(-t)$	$X(-f) = X^*(f)$, * is complex conjugate
5	Duality	$X(t)$	$x(-f)$
6	Frequency translation	$x(t)e^{j2\pi f_o t}$	$X(f - f_0)$
7	Modulation	$x(t)\cos(2\pi f_0 t)$	$\frac{1}{2}X(f - f_0) + \frac{1}{2}X(f + f_0)$
8	Differentiation	$\frac{d^n x(t)}{dt^n}$	$(j2\pi f)^n X(f)$
9	Integration	$\int_{-\infty}^{t} x(\tau)d\tau$	$(j2\pi f)^{-1}X(f) + \frac{1}{2}X(0)\delta(f)$

10	Convolution	$\int_{-\infty}^{\infty} x_1(t-\tau)x_2(\tau)d\tau = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau$	$X_1(f)X_2(f)$
11	Multiplication	$x_1(t)x_2(t)$	$\int_{-\infty}^{\infty} X_1(f-\xi)X_2(\xi)d\xi = \int_{-\infty}^{\infty} X_1(\xi)X_2(f-\xi)d\xi$

Some Trigonometry Formulas

$\sin x \cos y = \frac{1}{2}[\sin(x+y) + \sin(x-y)]$	$\cos x \sin y = \frac{1}{2}[\sin(x+y) - \sin(x-y)]$
$\sin x \sin y = \frac{1}{2}[\cos(x-y) - \cos(x+y)]$	$\cos x \cos y = \frac{1}{2}[\cos(x+y) + \cos(x-y)]$
$\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2}$	$\sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$
$\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2}$	$\cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}$
$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$	$\tan 2x = \frac{2\tan x}{1-\tan^2 x} = \frac{2\cot x}{\cot^2 x - 1} = \frac{2}{\cot x - \tan x}$
$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$	$\tan\frac{x}{2} = \pm\sqrt{\frac{1-\cos x}{1+\cos x}} = \frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$
$\tan\left(\frac{\pi}{2} - x\right) = \cot x$	

Q Function

$$Q(x) = \frac{1}{\sqrt{2\pi}}\int_x^{\infty} e^{\frac{-t^2}{2}}dt$$

Approximation of Q Function

$$Q(x) = \frac{1}{\sqrt{2\pi}}\int_x^{\infty} e^{\frac{-t^2}{2}}dt \approx \frac{1}{\sqrt{2\pi}x}e^{\frac{-x^2}{2}} \text{ for } x \gg 1$$

Joint Gaussian Density

The joint probability density function (pdf) of two correlated Gaussian random variables X and Y is given by

$$f_{xy}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} - 2\rho\frac{(x-\mu_X)(x-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(x-\mu_Y)^2}{\sigma_Y^2}\right]}$$

where $\mu_X = E[X]$, $\mu_Y = E[Y]$ are the mean values, $\sigma_X = \sqrt{E[(X-\mu_x)^2]}$ and $\sigma_Y = \sqrt{E\left[\left(Y-\mu_y\right)^2\right]}$ are the standard deviations of X and Y, respectively, and ρ is the correlation coefficient defined as $\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}$.

Problem I. (*Down-Conversion*) A receiver block diagram of a communication system is shown as in Figure 1, where the frequency responses of the bandpass filter and the lowpass filters can be expressed as

$$H_{\mathrm{BP}}(f) = \begin{cases} 1, & \text{if } |f-100| \le 10 \text{ Hz or } |f+100| \le 10 \text{ Hz}, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$H_{\mathrm{LP}}(f) = \begin{cases} 1, & \text{if } |f| \le 10 \text{ Hz}, \\ 0, & \text{otherwise}, \end{cases}$$

respectively. The received signal before the passband filter can be expressed as $x(t) = s(t) + n(t)$, where $s(t) = cos(210\ \pi t)$ is the modulated signal and $n(t)$ is the additive white Gaussian noise with power spectral density $S_N(f) = N_0/2$.

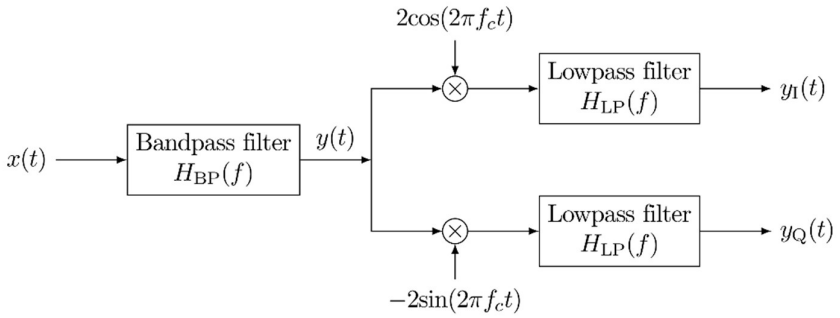


Figure 1: A receiver diagram

a) (4 *points*) Denote the output of the passband filter as $y(t) = \tilde{s}(t) + \tilde{n}(t)$, where $\tilde{s}(t)$ and $\tilde{n}(t)$ are the filtered signal and noise components, respectively. Find $\tilde{s}(t)$ and $E[|\tilde{n}(t)|^2]$.

$\tilde{S}(t) = S(t) = \cos(210\pi t)$

$E[|\tilde{n}(t)|^2] = \int N_0/2\ |H_{BP}(f)|^2 df$

$= N_0/2\ (1\times 20 + 1\times 20)$

$= 20N_0$

b) (4 *points*) Denote $y_I(t)$ and $y_Q(t)$ as the in-phase and the quadrature components, respectively, after down-conversion with $f_c = 100\ Hz$. Denote $y_I(t) = u_I(t) + n_I(t)$ and $y_Q(t) = u_Q(t) + n_Q(t)$, where $u_I(t)$ and $u_Q(t)$ are the in-phase and the quadrature baseband signals, and $n_I(t)$ and $n_Q(t)$ are the in-phase and the quadrature baseband noises, respectively. Find $u_I(t)$ and $E[|n_I(t)|^2]$.

$2\cos(200\pi t)\cos(210\pi t) = \cos(10\pi t) + \cos(410\pi t)$

$U_I(t) = \cos(10\pi t)$

$E[|n_I(t)|^2] = E[|\tilde{n}(t)|^2] = 20N_0$

c) (6 *points*) Find $\tilde{s}(t)$ and $E[|\tilde{n}(t)|^2]$ in a) and $u_I(t)$ and $E[|n_I(t)|^2]$ in b) if the frequency response of the bandpass filter is changed to

$$H_{\mathrm{BF}}(f) = \begin{cases} 1, & \textit{if } |f-100| \le 2 \text{ Hz or } |f+100| \le 2 \text{ Hz}, \\ 0, & \textit{otherwise}, \end{cases}$$

$\tilde{S}(t) = U_I(t) = 0$

$E[|\tilde{n}(t)|^2] = E[|n_I(t)|^2] = \int N_0/2\ |H_{BF}(f)|^2 df$

$= N_0/2 \times 8 = 4N_0$

Problem II. (*Analog Modulation*) Let $m(t)$ be a real message signal with $E[m^2(t)] = 1$, $m_p = max|m(t)| = 2$, and bandwidth $W = 10$ Hz. Denote $s_{\text{DSB}-}(t) = m(t)\cos(200\,\pi t)$. Figure 2 is a coherent demodulator to recover the message signal at the receiver.

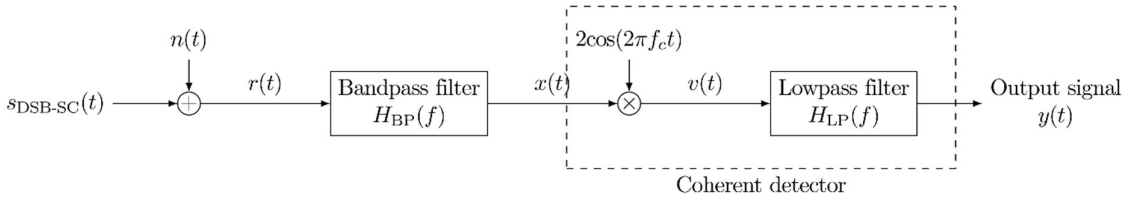


Figure 2. DSB-SC coherent demodulator

a) (4 *points*) Find f_c, the frequency response of bandpass filter $H_{\text{BP}}(f)$, and the frequency response of lowpass filter $H_{\text{LP}}(f)$ in Figure 2.

$f_c = 100$ Hz

$$H_{BP}(f) = \begin{cases} 1, & |f-100| \le 10 \text{ or } |f+100| \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

$$H_{LP}(f) = \begin{cases} 1, & |f| \le 10. \\ 0, & \text{otherwise.} \end{cases}$$

b) (3 *points*) If we ignore the channel noise in Figure 2, then the received signal at the receiver can be expressed as $r(t) = m(t)\cos(200\,\pi t)$. Let $s(t) = r(t) + A\cos(200\,\pi t) = m(t)\cos(200\pi t) + A\cos(200\,\pi t)$. Find the minimum A such that envelope demodulation can recover message signal $m(t)$ from $s(t)$ without distortion.

$$s(t) = (m(t) + A)\cos(200\pi t)$$

For envelope demodulation without distortion,

$$A \ge m_p \text{ or } A \ge 2$$

Minimum A will be 2.

c) (4 *points*) In b), if $s(t) = m(t)\cos(200\,\pi t) + 5\sin(200\,\pi t)$, can we still use envelope demodulation to recover $m(t)$ from $s(t)$ without distortion? can we use coherent demodulation to recover $m(t)$ from $s(t)$ without distortion?

* $$s(t) = m(t)\cos(200\pi t) + 5\sin(200\pi t)$$
$$= \sqrt{m^2(t) + 25}\cos(200\pi t + \theta(t))$$

The envelope is different from $m(t)$. Therefore, we cannot recover $m(t)$ using envelope demodulation.

* Coherent demodulation can still recover $m(t)$.

Problem III (*Digital Communication*) For the following baseband digital transmission system (Fig. 3.a), pulse waveform $g(t)$ is shown as in Figure 3.b. The linear time-invariant (LTI) filter in Figure 3.a is the matched filter.

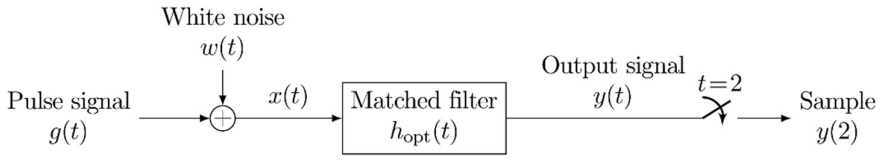


Figure 3.a: Matched filter for baseband digital transmission

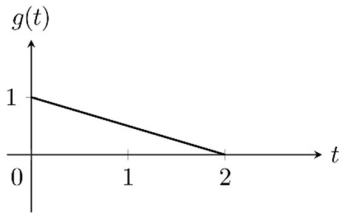
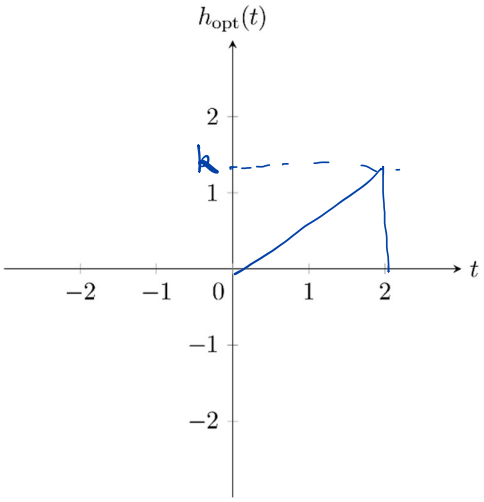


Figure 3.b: Pulse waveform

a) (4 *points*) Draw the impulse response of the matched filter

$h_{opt}(t) = k\,g(T_b - t)$
$= k\,g(2-t)$



b) (3 *points*) Let power spectral density (PSD) of the additive noise $w(t)$ be $S_W(f) = N_0/2 = 1/4$. Find the SNR at $y(2)$.

$S_w(f) = N_0/2 = 1/4, \quad N_0 = 1/2$

$E = \int g^2(t)\,dt = \int_0^2 \left(1 - \frac{1}{2}t\right)^2 dt$

$= \int_0^2 \left(\frac{1}{2}t\right)^2 dt = \frac{1}{4}\int_0^2 t^2\,dt$

$= \frac{1}{4}\cdot\frac{1}{3}\left(2^3 - 0\right) = 2/3$

$SNR = \frac{2E}{N_0} = \frac{2\times 2/3}{1/2} = 8/3$

Problem IV. (*Sampling and Quantization*) For (real) message $m(t)$, $P = E[m^2(t)] = 1$, $m_P = max|m(t)| = 2$, and bandwidth W=2 Hz.

a) (2 *points*) Find the minimal sampling frequency (rate) so that $m(t)$ can be accurately recovered from the samples.

$$f_s = 2 \times 2 = 4 \text{ Hz}$$

b) (*4 points*) Assume a uniform quantizer is used to the samples in a). Find the minimum number of bits per sample, n, so that the SNR at the quantizer output is no less than 40 dB.

$$SNR_o = \frac{3P}{m_p^2} 2^{2n}$$

$$= \frac{3 \times 1}{2^2} 2^{2n}$$

$$SNR_o(dB) = 10 \log_{10} SNR_o$$

$$= 6n - 1.25 \quad (dB)$$

$$6n - 1.25 \geq 40 \Rightarrow n \geq 6.88$$

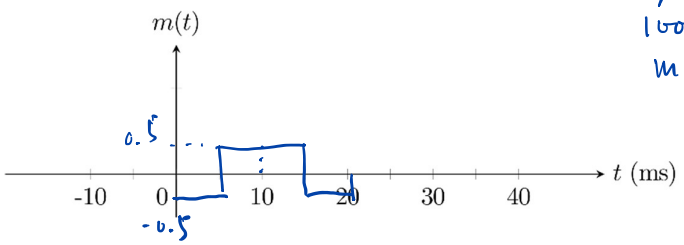
$$n = 7$$

Problem V. (*Digital and Analog Modulation*) For FSK, the modulated signal can be expressed as

$$s(t) = \begin{cases} A\cos(2\pi f_0 t), & \textit{if symbol "0" is transmitted,} \\ A\cos(2\pi f_1 t), & \textit{if symbol "1" is transmitted,} \end{cases}$$

where amplitude $A = 1$ volt, symbol duration $T = 5$ msec $(5 \times 10^{-3}$ sec), $f_0 = 950$ Hz, and $f_1 = 1050$ Hz. We can also express FSK signal into the form of frequency modulation (FM) as $s(t) = A\cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\, d\tau\right)$, where $k_f = 100$ and $f_c = 1000$ Hz.

a) (4 *points*) Draw the waveform of $m(t)$ if the transmitted symbol is $\{0, 1, 1, 0\}$.



symbol "0"
$1000 + 100\, m(t) = 950$
$m(t) = -0.5$
symbol "1"
$1000 + 100\, m(t) = 1050$
$m(t) = 0.5$

b) (2 *points*) Whether it is the minimum shift keying (MSK) or not?

$\Delta f = 1050 - 950 = 100$ Hz $\Rightarrow \Delta f = \frac{1}{2T}$
$T = 5 \times 10^{-3}$
It is MSK !

c) (*2 points*) What is its bandwidth of the FSK signal?

$$B_{MSK} = \frac{3}{2T} = \frac{3}{2 \times 5 \times 10^{-3}} = 300 \text{ Hz}$$

d) (*2 points*) What is the Carson bandwidth if $s(t)$ is regarded as a FM signal? [Hint: the bandwidth of a random sequence with symbol duration T is regarded as $W = 1/2T$.]

If regarding $s(t)$ as a FM signal,
then $\Delta f = 50$ Hz (different from in b)!).

$$B_T = 2(\Delta f + W) = 2\left(50 + \frac{1}{2 \times 5 \times 10^{-3}}\right)$$
$$= 300 \text{ Hz}$$

Problem VI. (*Information Theory*) In Figure 6, the discrete memoryless source (DMS) randomly generates symbol "0" or "1" every unit time with $\Pr(s = 0) = 0.8$ and $\Pr(s = 1) = 0.2$, the binary symmetric channel (BSC) transmits one symbol every unit time with probability of error $P_e = 0.2$, that is, $\Pr\{y = 1|x = 0\} = \Pr\{y = 0|x = 1\} = P_e = 0.2$.

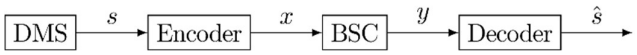


Figure 6. A communication system with a DMS and BSC

a) (4 *points*) Find the entropy of the DMS.

$$H(S) = h_p(p)$$
$$= -0.8\log_2 0.8 - 0.2\log_2 0.2$$
$$= 0.72 \text{ bit}$$

b) (4 *points*) Find the capacity of the BSC.

$$C = 1 - h_p(p)$$
$$= 1 - 0.72$$
$$= 0.28 \text{ bit}$$

c) (*4 points*) Are there an encoder and a decoder such that the probability of decoding error, $\Pr\{s \neq \hat{s}\}$, can be arbitrarily small? State the reason if the answer is "no"; design the encoder if the answer is "yes".

Since $H(S) > C$, there is no coding with arbitarily small decoding error.